SDC HACKATHON 2026

Phase 2: Final Project & Prototype Report

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1 November 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Final.pdf" together with your pitch deck and a working deployment link. Answers must comply with the word limits stated for each open-ended question. Section 3 requires concrete, verifiable evidence. Unsupported claims will not be credited by judges.

1. Team & Project Overview

Team Name	XH-Cat
Project / Product Name	FutureReady
Final Live Deployment URL	https://futures.grong.my
Codebase Repository URL	https://github.com/guirong0109/XH-Cat_MyFuture
Name (Leader)	YIP GUI RONG
Email & phone (Leader)	guirong05@gmail.com 011-61511138

2. Project Description

[Maximum 250 words]

FutureReady MY is a career-planning web app for Malaysian students and early-career adults choosing courses and jobs in an AI-changing labour market (SDG 4.4 and 8.6). A user builds a confirmed, structured profile from a PDF or Word CV or a short form: a rules reader extracts studies, jobs, skills, certificates and projects, and DeepSeek reads the CV alongside it, with every AI item tagged "Found by AI" for the user to check.

From the confirmed profile the app gives four tools. (1) Future Skills Radar: how much AI touches the tasks of the user's own jobs (Anthropic Economic Index, ILO WP140), AI research momentum (OpenAlex), and the latest AI news for the user's study field, found by DeepSeek Search every 3 hours and filtered by a relevance check. (2) Career matching: 1,812 Malaysian eMASCO roles with an explainable Skill Match (levels met divided by levels needed on official O*NET skills and knowledge), an Overall match that adds AI know-how, and Moderate or Rapid AI scenarios that change the outlook, never the match. (3) Study pathway: tick several roles and get the skills to build, each with an exact, checked course page, plus accredited Malaysian programmes. (4) An organisation view for HR teams and career centres.

Every number shows its source and date, and nothing is guessed. The app works in English, Bahasa Melayu and Chinese, for guests or with an account, and is live at futures.grong.my.

3. Upgraded Product & Technical Execution

3.1 Final System Architecture

Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.

Front end. A single-page web app in plain JavaScript ES modules (no framework): hash routing, one screen per module (profile, Radar, Career matching, Study pathway, organisation view), three.js for the landing intro, and all text in English, Bahasa Melayu and Chinese. Guest progress stays in the browser.

Back end. Node.js 24 with Express 5 serves the pages and a JSON REST API: /api/cv/extract, /api/skills, /api/occupations, /api/fields, /api/radar/*, /api/matching, /api/pathway, /api/org/*, /api/news and /api/auth/*. The matching, Radar, pathway and news logic sits in pure shared modules that run on both server and browser, so profile-based relevance is computed in the browser and the profile sent for matching is used for one answer and never stored or logged.

GenAI. The DeepSeek API is called in three places, each behind a guard: (a) the deepseek-flash chat model reads every uploaded CV in JSON mode next to the rules reader; an item is kept only if it quotes the CV word for word, and skill levels and codes never come from the model; (b) DeepSeek Search finds AI news for 10 study fields every 3 hours; (c) deepseek-flash keeps or drops each new news result (news about AI in that field). Hard caps of 10 searches and 10 checks per run and 80 a day are counted in memory and on disk before each call. Career matching itself uses no language model.

Data and database. Official datasets are bundled with SHA-256 manifests and rebuilt by Python pipeline scripts: O*NET 31.0, eMASCO (MASCO 2020), ISCED-F 2013 with the Statistics Canada and O*NET CIP crosswalks, the BLS SOC/ISCO code chain, ILO WP140, the Anthropic Economic Index, OpenAlex, DOSM job-ad data and the UN SDG targets. Accounts, sessions, saved profiles and organisations are stored in SQLite (Node's built-in node:sqlite) with the same interface as Cloudflare D1; the news lists are kept in a JSON store.

Hosting. The app runs on the team's computer and is published through a Cloudflare Tunnel at https://futures.grong.my (Cloudflare DNS, TLS and caching, no open router ports). A stable public copy and a local test copy run side by side; a change goes public only after it is tested and pushed.

Quality. 1,765 automated Vitest tests and 49 Python tests; every feature went from an approved spec and architecture diagram to a test-first plan, with a step-by-step build record in the repository.

3.2 Implementation & Hackathon Deliverables

Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.

[Maximum 300 words]

Built and deployed in the final round (first commit 29 Oct 2026, 482 commits on main): the CV and structured profile module with DeepSeek reading; the Future Skills Radar; career matching over 1,812 eMASCO roles through the official eMASCO, ISCO-08, SOC and O*NET code chain; the study pathway; the organisation view; AI news inside the Radar; accounts; and the live site futures.grong.my.

How we improved on the Phase 1 design:

(1) Phase 1 compared embedding models for matching. We replaced similarity scores with arithmetic on official O*NET levels and official code tables, so every match can be explained on screen ("you meet 9.2 of 12.05 needed levels, 76%").

(2) The pathway grew from 38 app skills to all 68 O*NET skill and knowledge items, each with an exact, checked course page, plus accredited programmes for 99 of 124 fields of study.

(3) The rules-only CV reader missed items, so DeepSeek now reads every CV too, with a word-for-word evidence check and the user confirming each AI item.

(4) The first real news run kept preprint listings and homework pages, so we added a DeepSeek relevance check; it now drops 6 to 34 non-news results per field each refresh.

(5) Two whole-branch code reviews found a cost-cap bypass and a crash path; both were fixed with tests.

(6) We moved hosting to a Cloudflare Tunnel with local SQLite (no paid cloud database) and removed two placeholder tabs to keep the journey focused.

4. Environmental & SDG Impact Verification

Validate the real-world impact of your built solution. Show concrete data that validate your environmental or social value proposition, compared against the baseline and KPI defined in previous phase submission.

*The data and evidence you provide in this section will be used to assess **SDG Impact, Technical Execution and Innovation scores.** Vague claims or missing verification will weaken your score across all three categories.*

Metric Category	Your Response
Target SDG Metric Achievement *Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers* *e.g., "Achieved a 24.5% reduction in computational execution overhead against the 20% initial target."*	Phase 1 KPI: share of pathway steps that give an exact, checked official course page. Phase 1 baseline: 82% (31 of 38 skills, 24 Oct 2026). Phase 1 target: at least 90%. Phase 2 prototype: 100%. All 68 O*NET skill and knowledge items that the pathway uses as steps have at least one exact, checked course page (125 course picks, checked 31 Oct 2026), beating the 90% target by 10 points. Accredited Malaysian programme pages also cover 99 of 124 ISCED-F fields of study (79.8%). Note: the base grew from 38 app skills to the 68 O*NET items that Skill Match uses. The planned pilot with at least 20 students (70% finishing in under 10 minutes) has not run yet, so no result is claimed for it.
Verification Evidence Method *How was this data verified?* *e.g., codebase algorithmic profiling logs, hardware power-meter simulation readings, or database transaction counters.*	Counted from the repository data, not estimated: data/pathway/picks.json holds every course and programme pick with its exact link and the date it was checked, under a SHA-256 manifest; the server refuses to start with a bad pick. docs/pathway/fill-report.md lists every gap with the sites searched. CV extraction was measured on 16 synthetic CVs (docs/eval-results.md): precision 1.0 on every field, skill recall 0.89, 0 wrong items. A live run on futures.grong.my refreshed all 10 news fields with no error. 1,765 automated tests pass.
Socio-Economic / Civic Value *How does your deployed application translate into a scalable, real-world solution capable of being supported by civic or corporate partners?*	Malaysian youth unemployment is 10.2% (OpenDOSM, June 2026) and only 26% of professionals have formal AI training. FutureReady turns official but scattered data (eMASCO, MQA programmes, DOSM job ads, O*NET) into one free, explainable plan in three languages. Career

Metric Category	Your Response
	centres and TVET colleges can use the organisation view to see a cohort's skill gaps and plan reskilling; HR teams can plan for target roles. The tool suits support from universities, HRD Corp, MOHR and employer networks, because it shows sources rather than opinions.
Deployment & Commercial Feasibility *e.g., modeled as an API service plugin designed for easy integration into existing corporate or institutional systems.*	Built as JSON API services behind one web app, so an institution can embed the matching, pathway or news endpoints in its own portal. Running cost is low: about US$4 to US$6 a month for DeepSeek (news) plus about US$0.002 to US$0.004 per CV read, on free Cloudflare DNS and Tunnel. The same storage interface runs on Cloudflare D1, so it can move to managed cloud hosting without code changes. A paid organisation view for HR and career centres, with a free student app, is the planned model.

4.1 Mathematical Proof / Code Logic Snippet (Optional, but holds bonus marks)

Paste the specific code section, mathematical formula, simulation log output, or calculation logic that proves the impact claims made above.

This proof is graded across three categories at once: ***Impact and Feasibility*** *(does the math support your claim),* ***Technical Execution*** *(does it reflect what you actually built), as well as* ***Innovation and Creativity*** *(is the approach original).] Do not leave this section generic or copied from Phase 1.*

Skill Match (shared/matching/fit.js), the core claim behind every match:

Skill Match = sum over items of min(you, needed) / sum over items of needed, over O*NET skills and knowledge with importance of at least 3.0. Your level: basic 2, good 4, expert 6 on the O*NET 0 to 7 scale.

let met = 0, needed = 0

for (const r of rows) { met += Math.min(r.you, r.needed); needed += r.needed }

return { met, needed, percent: needed > 0 ? Math.round((met / needed) * 100) : null }

Worked example (an automated test): needed 12.05, met 9.2, so 9.2 / 12.05 = 0.763, shown as 76%.

Overall match (shared/matching/overall.js):

Overall = Skill Match x (1 - AI touch) + AI know-how x AI touch, where AI touch is the observed task share (Anthropic Economic Index, Moderate) or the ILO exposure score (Rapid). It never changes the Skill Match.

Pathway KPI (data/pathway/picks.json):

covered = items.filter(k => picks.some(p => p.for === k && p.link.startsWith("https://"))).length

Result on main: covered = 68, items = 68, so 100% (Phase 1: 31 / 38 = 82%; target 90%).

5. Market Viability & Future Scalability Plan

SDC aims to encourage solutions that transition from innovations into market-ready applications. If given access to corporate funding, industry grants, or partnership opportunities (e.g., through SDC's Rotary network or joint research initiatives at XMUM), how would your team scale this prototype to process real-world data at volume?

[Maximum 250 words]

With partner funding we would scale in four steps. (1) Data at volume: replace bundled snapshots with live feeds from DOSM job ads, HRD Corp course catalogues, MQA programmes and the O*NET Web Services API, refreshed automatically with source dates shown, as the Radar and news already do.

(2) Pilots: run the planned study with at least 20 students, then cohorts at XMUM, MMU career centres and TVET colleges, measuring time to a pathway and whether students know which skill to learn first.

(3) Infrastructure: move from the team's computer to managed hosting (Cloudflare D1 and R2 already supported by the code), add AI news in Bahasa Melayu and Chinese, and review data handling against Malaysia's PDPA before real users upload CVs.

(4) Sustainability: keep the student app free and offer the organisation view to universities, TVET providers and employers, including Rotary network companies, as a paid service for cohort skill-gap planning. Running cost stays low because matching is arithmetic on public data and the AI calls are capped.

6. Final Submission Checklist

Tick all boxes that apply before submitting.

- [x] The application is fully compiled, live, and accessible via the deployment link provided in Section 1.
- [x] The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.
- [x] The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.
- [x] The final pitch deck has been exported and saved as a flattened PDF file.
- [x] All fields in this template have been completed and word limits respected.

7. IP & Originality Declaration

7.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No ☒ No change from Phase 1 declaration

If yes, briefly describe what was pre-existing and which team member(s) owned it:

7.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No ☐ No change from Phase 1 declaration

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Node.js 24.16	code (runtime)	MIT
Express 5.3.0	code (library)	MIT
pdfjs-dist 6.4.299	code (library)	Apache-2.0
three.js 0.186.1	code (library)	MIT
Vitest 5.0.5	code (test tool)	MIT
Geist, Geist Mono, Instrument Serif (Google Fonts)	code (fonts)	SIL Open Font License 1.1
DeepSeek deepseek-flash (chat API)	model	DeepSeek API terms (paid)
DeepSeek Search API	model / API	DeepSeek API terms (paid)
O*NET 31.0 database	dataset	CC BY 4.0 (USDOL/ETA)
eMASCO, MASCO 2020 (MOHR)	dataset	Public government reference

ISCED-F 2013, Statistics Canada concordance, O*NET CIP crosswalk	dataset	Statistics Canada Open Licence; CC BY 4.0
BLS SOC 2018/2010 and SOC/ISCO-08 crosswalks	dataset	US government work (public domain)
ILO Working Paper 140 GenAI exposure scores	dataset	CC BY 4.0
Anthropic Economic Index (2026-06-26)	dataset	CC-BY
OpenAlex research counts	dataset	CC0
DOSM Job Market Insights	dataset	Public DOSM release
UN SDG goals and targets (UNSD SDG API)	dataset	UN material

7.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No ☒ No change from Phase 1 declaration

If yes, please describe the nature of the agreement and which member it concerns:

8. Declaration

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.

YIP

1 November 2026